

Case Report

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**Acute Myocardial Infarction in a Young Woman: Unexpected Findings of a
Coronary Occlusion**

Shuquan Zhao^{1,†}, Longlong Zhu^{1,†}, Lopsong Tinzin¹, Fang Huang¹, Longda Ma, Yiwu Zhou^{1,*}

¹ Department of Forensic Medicine, Tongji Medical College, Huazhong University of Science and Technology, No. 13 Hangkong Road, Wuhan, 430030, PR China

* Corresponding author.

[†]Shuquan Zhao and Longlong Zhu contributed equally.

E-mail addresses:

fyzs123@126.com (S. Zhao),

yuexiashaungzhuo@126.com (L. Zhu),

m13164135772@163.com (L. TinZing),

rita_hf@126.com (F. Huang),

2015754397@qq.com (L. Ma)

zhouyiwu@outlook.com (Y. Zhou), Tel 13707160935.

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Conflict of Interest Statement

The authors declare that they have no conflict of interest.

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Abstract

In clinical and forensic practice, the cause of death is often attributed to acute

myocardial infarction, among which the coronary atherosclerosis being the Captain of the Men of Death. However, other reasons such as coronary septic embolization with neutrophilic granulocyte myocarditis although rare, can also cause sudden unexpected death. This paper reports a case with this rare cause—a 21-year-old woman diagnosed with "acute gastroenteritis" who died 4 days later. A forensic autopsy revealed an inflammatory polypous embolic located at 1.0 cm from the left anterior descending branch (LAD) with severe neutrophilic granulocyte myocarditis, which resulted in embolic at the opening of the left main coronary artery, acute myocardial infarction and eventually leading to her death. Histopathological examination showed large amounts of neutrophilic granulocyte infiltration in the arterial layer forming the septic embolic and eventually resulting in coronary occlusion. To find the real cause of septic embolic, myocarditis, bacterial, fungal, protozoan and virus detection was performed through RT-PCR, with negative findings. Septic embolic leading coronary occlusion in left main coronary artery and LAD is rarely reported in forensic practice, we hope this report can pave the way on understanding this rare disease to make correct diagnosis in medical practice.

Keywords: forensic science; forensic pathology; Myocarditis; Coronary embolic; Myocardial infarction ; Myocardial infarction

Myocardial infarction (MI), a common presentation of ischemic heart disease/coronary artery disease, is the leading cause of sudden death worldwide (1). Coronary heart disease (CHD) is the most common causative disease underlying coronary occlusion and it can lead to myocardial ischemia and necrosis. Although the mortality rate of CHD has declined significantly in developed countries over the past few decades, the disease is still the leading killer of adults in the United States (2). MI is most common among the elderly, but there are also reports of death from MI in young adults (3). In fact, sudden cardiac death (usually caused by CHD) is the most common cause of sudden death in adults under 40 years old (4–5). In this paper, we present the sudden death of 21-year-old young woman whose cause of death is unknown. A septic embolic blockage at 1.0 cm from LAD and the opening of the left main coronary artery and severe neutrophilic granulocyte myocarditis were observed in forensic autopsy, resulting in extensive myocardial infarction. No atherosclerotic plaques or other lesions other than large amounts of neutrophilic granulocyte infiltration were found in media and adventitia of LAD where the septic embolic was found.

Case History

The deceased was a 21-year-old woman who went to the clinic for abdominal pain at 10:00 a.m. She had no significant past medical history and no history of intravenous drug use or heart diseases. Physical examination (PE): T: 36.5°C, P: 76 beats/min, regular rate and rhythm, no murmurs, rubs or gallops. She was diagnosed with acute gastroenteritis at a local hospital and was given intravenous infusion of 250 ml 0.9% sodium chloride and 40 mg omeprazole. During the infusion, the patient experienced no relief for abdominal pain and developed new symptoms of dizziness,

chest tightness, and vomiting. At 3:00 p.m., when the patient returned home after the infusion, she suddenly turned pale, lost consciousness, and stopped breathing. She was sent to the ICU and underwent tracheal intubation. PE: T: 36.6°C, P: 140 beats / min, R: 8 breaths / min, BP: 80/40 mmHg. ECG indicated sinus tachycardia, paroxysmal tachycardia, ST-T pattern for acute ischemic injury and abnormal q wave. Hypersensitive troponin T: 1180.00 ng/L; B-type natriuretic peptide precursor (NTproBNP): 206.10 pg/ml; aspartate aminotransferase: 207 U/L; lactate dehydrogenase: 592 U/L; and creatine kinase: 377 U/L. At 5:40 p.m. on the same day, treatments were given for mild hypothermia, reduction of brain edema, and the protection of brain cells. The patient developed multiple ventricular fibrillation. Although defibrillation and cardiopulmonary resuscitation have been performed, the patient died four days later.

Autopsy findings

Four days after the patient's death, a forensic autopsy was performed and the following macroscopic findings were made. No obvious abnormalities were found in bilateral external auditory area or in the oral mucosa or teeth. There was evidence of medical intervention and no evidence of pre-existing abnormalities or injuries.

There were hemorrhage on the left margin of the heart, sized 5.0 cm × 3.0 cm, on left endocardium and a hemorrhage sized 3.0 cm × 2.5 cm on the right atrium (Fig. 1a). The color of the posterior interventricular groove was fresh, dark pink which indicated septal infarct (Fig. 1b). There was no evidence of atherosclerosis or arterial abnormality predating the disease being illustrated. A polypous, rough-surfaced, grayish vegetation, size 0.5 cm × 0.4 cm, attached at the opening of the coronary artery (Fig. 2a, b). Heart valves showed no abnormalities.

Left ventricular and interventricular septum myocardial cells showed extensive

wave-like changes. Some cells were brittle and disintegrated with myoglobin lysis, contraction band necrosis, and significant nuclear pyknosis. A large amount of neutrophil infiltrations was observed in the myocardial interstitial space (Fig. 3).

The vegetation at the opening of the coronary artery showed as a thrombus-like structure composed of neutrophils, mononuclear lymphocytes, fibroblasts, collagen fibers, platelets, and cellulose. The base of the vegetation was attached to the aortic wall (Fig. 2c). No atherosclerosis was observed in any main branch of coronary artery.

A mixed thrombus-like blockage with a diameter of 0.3 cm was seen at 1.0 cm from the left anterior descending branch (Fig. 4a). Under the microscope, this thrombus was found to be composed of coral-like platelet trabecular cells and neutrophils adhering to the trabecular edges. Some of these cells were necrotic and disintegrated, and a large amount of red blood cells and net-structured cellulose were observed. Quantities of inflammatory cells, tissue necrosis and hemorrhage were seen in media and adventitia of the LAD, the internal elastic membrane was discontinuous (Fig. 4b). The blockage was serially sectioned and paralleled with VEG staining, Masson staining revealed thrombus and discontinuous internal elastic membrane (Fig. 4c, d). The blockage was serially sectioned and paralleled with PAS staining, Gram staining with negative results. No further pathological changes other than edema and ischemia were observed in other organs. As no bacterial colonies were observed, we conducted molecular examination to find the possible bacterial or virus.

Laboratory examinations

Bacterial detection

Rapid sequencing to identify the pathogenic species was performed according to the NGS assay manual (BGI-Wuhan, China) (6,7). Briefly, genomic DNA were

extracted from paraffin-embedded sections of vegetation areas of the heart using MagPure FFPE DNA LQ Kit (Magen, China) according to the manufacturer's instructions. DNA libraries were constructed after end repairing, adapter ligation and PCR. Sequencing was performed by using the BGISEQ-100 platform after quality control by 2100 bioanalyzer system (Agilent Technologies, Santa Clara, CA, USA). Human sequences were removed and the remaining sequence data were aligned to bacterial, viral, fungal, and protozoan databases which includes 2406 bacteria, 1424 viruses, 199 fungi, 135 parasites, 5 tuberculosis pathogens and 41 mycoplasma/chlamydia in *Mycobacterium tuberculosis* complex.

The detection results of experimental samples and the control samples show: cardiac vegetations mainly detected non-pathogenic Gram-negative bacteria (such as: *Caulobacter*, *Sphingopyxis* bacteria), did not detect Gram-positive bacteria.

Virus detection

Ten tissue samples which included brain, spinal, heart, liver, spleen, lung, kidney, stomach, sublingual gland and jejunum were taken and sent to Wuhan Institute of Virology, Chinese Academy of Sciences (CRS) for virus detection. Since about half of viral myocarditis are associated with enterovirus infection, we used universal primers for detection of three enteroviruses. At the same time, two kinds of universal primers of CVB3 were used to detect Coxsackie virus B3, which was clinically reported to be closely related to viral myocarditis, and we detected five kinds of DNA viruses (adenoviruses, HSV-I; HSV-II; CMV and EBV) that may cause viral myocarditis. RT-PCR and DNA virus detection results showed that the virus detection was negative.

Discussion

In this case, with autopsy, it was not difficult to find the cause of the death, a

septic embolic located at the LAD and opening of Left main coronary artery. This septic embolic with myocarditis caused the acute myocardial infarction, heart failure, and eventually death. Histopathological examination showed that the septic embolic mainly consisted of large numbers of neutrophils, mononuclear lymphocytes, fibroblasts, and collagen fibers. The septic embolic located at the LAD, formed a thrombus and blocked the lumen, thus causing acute myocardial infarction.

Coronary embolism is a rare disease and most occurs in LAD, LCX and RCA, the Left main coronary artery is rarely involved (8). It is a known complication of infective endocarditis especially those involving the aortic valve, other possible etiologies include atrial fibrillation, valve replacement, venous thrombosis, it can also occur after cardiac surgery or catheterization in rare circumstances (9). Coronary embolism is a rare cause of acute myocardial infarction and accounts for a significant percentage in those who died of acute myocardial infarction without atherosclerosis. It is reported that coronary embolism had been diagnosed on autopsy about 13% of acute myocardial infarction deaths (10).

Only a few conditions have been shown to cause septic embolic located at LAD and the opening of the coronary artery. Infective endocarditis (IE) is the most common cause of coronary artery vegetation. Its clinical symptoms are mild, variable, and even non-specific (fever, night sweats, joint pain, anorexia), and they are often overlooked or missed in diagnoses made in clinical settings (8). D. Geisenberger *et al.* reported a similar case (9): a deceased young man with a history of drug abuse. Valve vegetation with a serrated pattern, degeneration, ulceration, and attached inflammatory vegetation were accidentally discovered after the patient's death. Unlike in this previous case, our patient had only an isolated vegetation at the opening of the left main branch. Her heart valves were intact and no other lesions were observed. As

a mixed thrombus, the isolated vegetation was composed of many fibroblasts and collagen fibers. No bacterial clone was detected by gram staining, which suggested an aseptic chronic inflammatory vegetation.

Intravenous drug use (IVDAS) may be one cause of bacterial growth (10). Although several vascular punctures were found on the body of this young woman, they were results of recent medical care. No history of drug abuse, intravenous or otherwise, was recorded for this patient. Gene amplification and sequencing were performed in order to further explore the cause of the vegetation. The results were negative for pathogenic bacteria for IE, such as dominant staphylococcus (65.6%) and streptococcus (25.6%) (11). PCR test results were also negative for viruses, so we ruled out viral myocarditis. The other detected bacteria were confirmed to be post-mortem contamination. Based on these findings, the reported case is easily to conclude the exact cause of this septic embolic was the infiltration in the LAD. The embolism at the opening of the left main coronary artery maybe the result of the septic embolic in LAD.

Currently, there are an increasing number of cases of acute myocardial infarction in young people, even in those without any significant medical history. This current reported case could be used as a reference case in clinical practice, when a young female patient is found to have a coronary event. The location of the embolic in this case should also draw attention to areas besides the heart valves, so allowing patients to be treated promptly and effectively. This case could also be used as a reference and provide useful information for forensic practice. In addition, to make the most accurate diagnosis possible, it is also necessary to conduct histopathological examination, pathogen screening, and obtain the medical history of the patient.

REFERENCES

1. Sahoo S, Losordo DW. Exosomes and cardiac repair after myocardial infarction. *Circ Res*, 2014, 114(2):333-44.
2. Roger VL, Go AS, Lloyd-Jones DM, Adams RJ, Berry JD, Brown TM, *et al.* Heart disease and stroke statistics-2011 update a report from the American Heart Association. *Circulation*. 2011;123(4): e18–e209.
3. Takegoshi T, Haba T, Hirai J, Saga T, Kitoh C, Mabuchi H. A case of hyperLp(a) aemia, associated with systemic lupus erythematosus, suffering from myocardial infarction and cerebral infarction. *Jpn J Med*, 1990,29(1):77-84.
4. Arzamendi D, Benito B, Tizon-Marcos H, Flores J, Tanguay JF, Ly H, *et al.* Increase in sudden death from coronary artery disease in young adults. *Am Heart J*, 2011, 161(3):574-80.
5. Drory Y, Turetz Y, Hiss Y, Lev B, Kramer MR. Sudden unexpected death in persons <40 years of age. *Am J Cardiol*, 1991, 68(13):1388-92.
6. Ye M, Wei W, Yang Z, Li Y, Cheng S, Wang K, *et al.* Rapid diagnosis of *Propionibacterium acnes* infection in patient with hyperpyrexia after hematopoietic stem cell transplantation by next-generation sequencing: a case report. *BMC Infect Dis*, 2015, 16:5.
7. Yao M, Zhou J, Zhu Y, Zhang Y, Lv X, Sun R, *et al.* Detection of *Listeria monocytogenes* in CSF from Three Patients with Meningoencephalitis by Next-Generation Sequencing. *J Clin Neurol*, 2016, 12(4):446-51.

8. Vendittelli PS, Botros B, Rosman HS, Govindaraju V, Zaitoun A, Marroush TS. Coronary Artery Embolism: Two Case Reports and a Review of the Literature. *Am J Med Sci.* 2019;357(4):333-7.
9. Shamsham F, Safi AM, Pomerenko I, Saliccioli L, Feit A, Clark LT, *et al.* Fatal left main coronary artery embolism from aortic valve endocarditis following cardiac catheterization. *Catheter Cardiovasc Interv*, 2000, 50(1):74-7.
10. Kotooka N, Otsuka Y, Yasuda S, Morii I, Kawamura A, Miyazaki S. Three Cases of Acute Myocardial Infarction Due to Coronary Embolism. *Jap Heart J*, 2004, 45(5):861-6.
11. Bayer AS, Bolger AF, Taubert KA, Wilson W, Steckelberg J, Karchmer AW, *et al.* Diagnosis and management of infective endocarditis and its complications. *Circulation*, 1998, 98(25):2936-48.
12. Geisenberger D, Huppertz LM, Büchsel M, Kramer L, Pollak S, Große Perdekamp M. Non-traumatic subdural hematoma secondary to septic brain embolism: A rare cause of unexpected death in a drug addict suffering from undiagnosed bacterial endocarditis. *Forensic Sci Int*, 2015, 257:e1-e5.
13. Moreillon P, Que YA, Infective endocarditis, *Lancet*, 2004, 363 (9403) :139–49.
14. Mathew J, Addai T, Anand A, Morrobel A, Maheshwari P, Freels S. Clinical features, site of involvement, bacteriologic findings, and outcome of infective endocarditis in intravenous drug users. *Arch Intern Med*, 1995,155(15): 1641–8.

Figure legends :

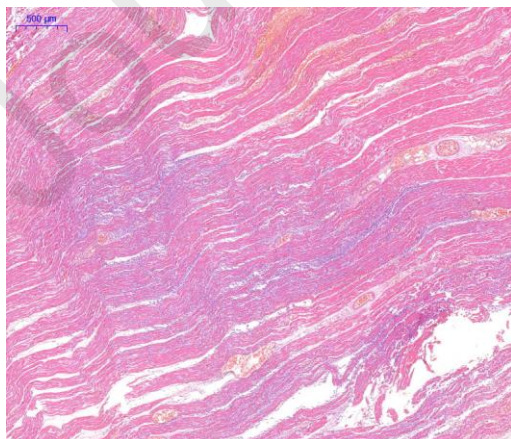
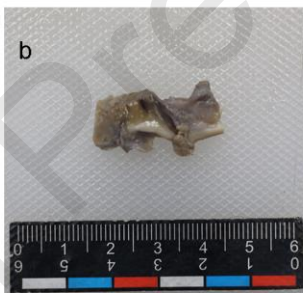
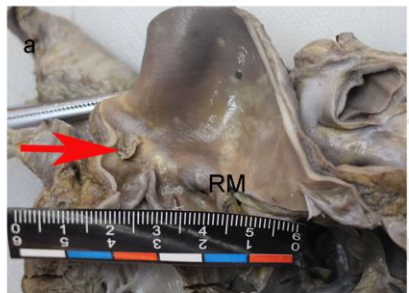
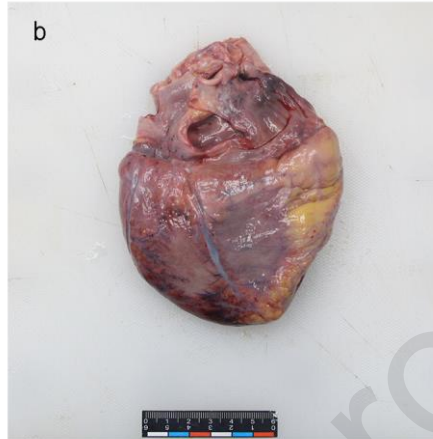
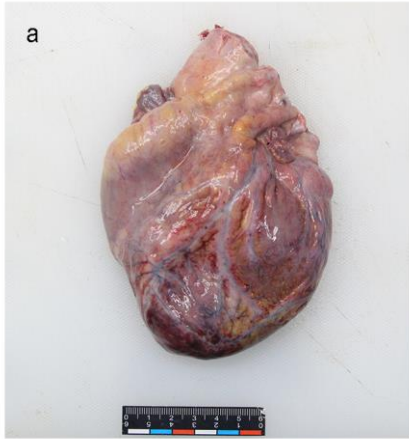
Fig.1. Hemorrhage on left endocardium , hemorrhage on right atrium, the color of the posterior interventricular groove was fresh, dark pink which indicated septal infarct.

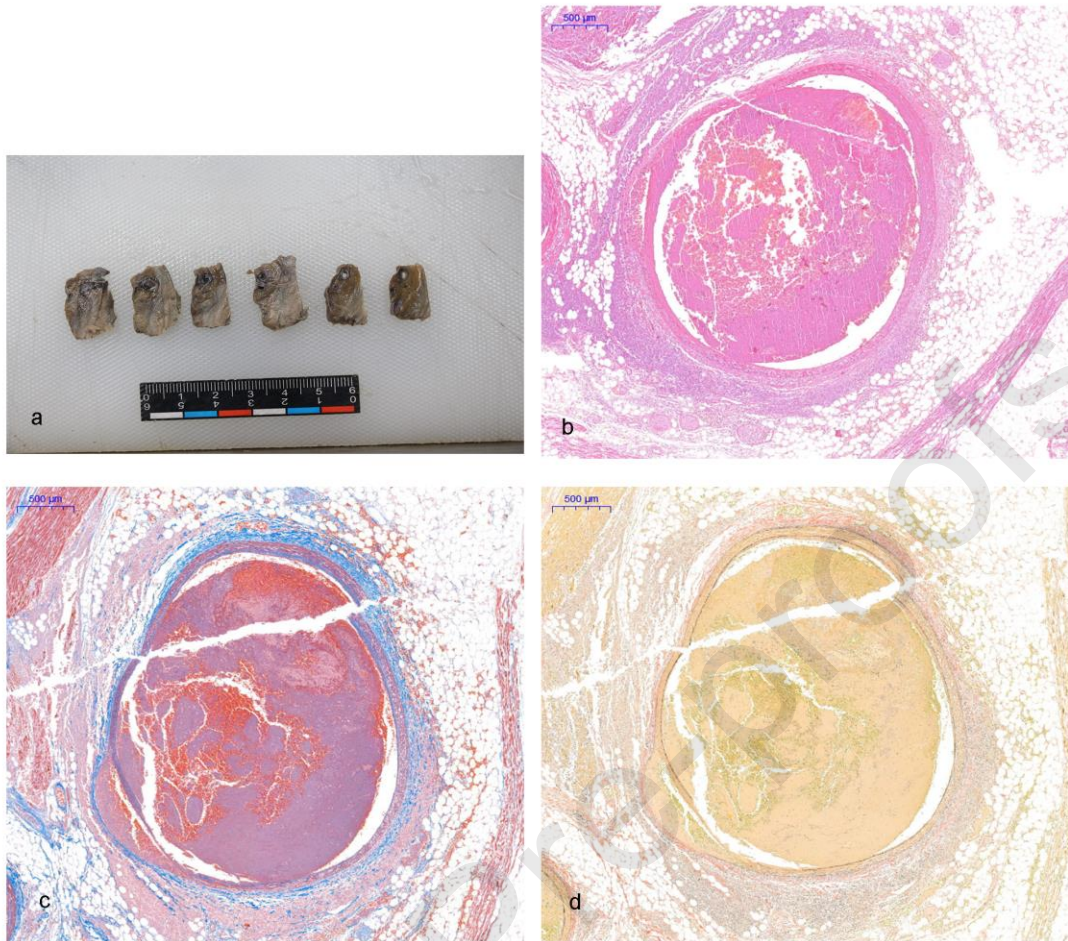
Fig.2. Septic embolism at the opening of the left main coronary artery. a, b): A polypous, rough-surfaced, grayish vegetation, size 0.5 cm × 0.4 cm, attached at the opening of the coronary artery; c): The embolic at the opening of the coronary artery showed as a thrombus-like structure composed of neutrophils, mononuclear lymphocytes, fibroblasts, collagen fibers, platelets, and collagen (HE&20×).

Fig.3. Myocardium cells showed significant nuclear pyknosis with large amounts of neutrophil, lymphocyte infiltrations and fresh blood in the myocardial intestinal space (HE&20×).

Fig.4. The septic embolism in the LAD. a): A mixed thrombus-like blockage the LAD. The structure of the thrombus and the inflammation in media and adventitia of LAD,

b): HE&20×, c): VEG&20×, D): Masson&20×.





Highlights

1. We report a case of sudden death due to a coronary occlusion caused by septic embolism with myocarditis.
2. The cause of death was acute myocardial infarction caused by coronary embolic in LAD and the opening of left main coronary artery, which was diagnosed via an autopsy and pathological examination.
3. Laboratory examinations showed negative bacterial detection and virus detection of heart vegetation.
4. Coronary embolism is a rare cause of acute myocardial infarction and account for

a significant percentage in these died of acute myocardial infraction without atherosclerosis.